Two-Copy Symmetry-Sector Discrimination Beyond Matched Block Purities

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Abstract

Subsystem purity need not capture all information available from a two-copy Bell measurement. We give a finite, explicitly promised discrimination problem in which a periodic cluster representative and a time-reversal-trivial entangled dimer representative have equal purities for every union of two-qubit blocks. Their complete joint block-SWAP laws consequently coincide. Resolving the four tensor-square sectors of blockwise SO(4) instead gives exact equal-prior success probability 3/4 using two copies of an eight-qubit state. A rational weak-duality certificate bounds all measurements with positive partial transpose across the copy cut below 0.669068, including all sequential single-copy strategies with classical memory and no quantum communication between copies. The sector measurement is invariant to an arbitrary, fixed product of unknown symmetry-preserving block rotations. A four-dimensional transfer representation evaluates its likelihood in linear time in the number of blocks. On 400 held-out physical states, the sector classifier reaches 97.075% accuracy at eight source copies. A model-aware Monte Carlo single-copy reference, added as a post-hoc falsifier, reaches 81.600% at that budget. We also give an exact finite Clifford twirl for a specified readout-noise model and account for its added gates. These results concern two symmetry-preserving orbits of fixed points, not arbitrary states in either phase. The theorem concerns two total copies, while the longer-budget comparison is empirical; neither establishes an asymptotic sample-complexity or physical-hardware advantage.

1 Introduction

Quantum phase recognition asks for a label from measurements of prepared many-body states. The measurement access matters: a joint measurement on two copies can retain information absent from a record of separate-copy measurements. Equally, an apparent advantage can result from discarding useful classical information. Quantum convolutional neural networks (QCNNs) and their experimental demonstrations provide established phase classifiers [1–3]; strong local hypothesis tests and classical surrogates can succeed on conventional benchmarks [4, 5]. An informative comparison must therefore specify the physical promise, retain the full relevant record, and compare equal resources.

We study a narrower question: can two phase representatives remain indistinguishable by every block-purity statistic while an equally shallow refinement of the same Bell readout distinguishes their symmetry sectors? We construct such a pair. The two representatives have different time-reversal phase ancestry, yet their purities agree for every union of declared blocks. Ordinary block-SWAP outcomes lose the distinguishing information. The four sectors of the blockwise SO(4) tensor-square representation retain it.

The main result is finite and exact. With four two-qubit blocks and equal class priors, one transversal Bell measurement of two copies, followed by sector decoding, attains success probability 3/4. Twirling makes this optimal for the nuisance-averaged two-copy problem. A rational certificate puts every positive-partial-transpose (PPT) measurement across the copy partition below 0.67. Thus no

strategy that measures each copy separately and carries only classical information between them can match the two-copy measurement at this total budget. The statement permits arbitrary entangling measurements *within* a copy; the restriction is between copies.

Bell sampling itself is established [6, 7], as are learning advantages from coherent processing of experimental copies [8]. Nonlocal symmetry-protected topological (SPT) order parameters and randomized-measurement access to many-body topological invariants are also established [9, 10]. Restricted-measurement discrimination and PPT bounds are standard tools [11]; Mele and Lami use a closely related tensor-power, four-sector twirl and PPT convex reduction for different data-hiding states [12]. Real Clifford orthogonal designs are likewise established [13]. The contribution here is the explicit matched-purity discrimination problem, its native sector likelihood, and a reproducible rational separation for its two-copy resource model. These ingredients do not establish a general quantum phase-learning advantage or a new principle of group-symmetric PPT optimization.

2 Physical promise and measurement access

Take a periodic chain of $N = 2M$ qubits, $M \geq 3$, with known blocks $B_j = (2j, 2j+1)$ and indices modulo N. Let

$$\Theta = \left(\prod_{i=0}^{N-1} X_i \right) K, \tag{1}$$

where K is complex conjugation in the computational basis. This is the time-reversal representation used for the cluster–Ising setting in Ref. [3].

The SPT representative $|C\rangle$ is the periodic graph-cluster state, stabilized by $Z_{i-1}X_iZ_{i+1}$. The opposite cluster-stabilizer sign has the same Bell probabilities and does not change the construction. The second representative is

$$|D\rangle = \bigotimes_{j=0}^{M-1} |\Phi^+\rangle_{2j+1,\,2j+2}, \qquad |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{2}$$

Its interblock entanglement does not make it SPT under Eq. (1). On each disjoint dimer bond,

$$\exp(i\pi Z \otimes Y/4)|{+}{+}\rangle = |\Phi^+\rangle. \tag{3}$$

The continuous gate path $\exp(itZ \otimes Y)$ commutes with the antiunitary $(X \otimes X)K$: both i and $Z \otimes Y$ change sign. Applying these gates in one layer connects $|D\rangle$ to the trivial product representative through symmetry-preserving local unitaries. The cluster representative has the nontrivial time-reversal ancestry [3]. Appendix A independently verifies the opposite virtual time-reversal indices of the two representatives. The finite-system labels refer to these specified, uniformly gapped parent-state families.

For each block define

$$V = (HS) \otimes (HS), \quad S = \mathrm{diag}(1, i), \qquad U_j = VO_jV^\dagger, \quad O_j \in \mathrm{SO}(4), \tag{4}$$

with H the Hadamard gate. Since $VV^{\mathsf{T}} = X \otimes X$, V changes from a real basis to the physical time-reversal basis, and U_j commutes with block time reversal. Each O_j has a path to the identity. Conjugating a representative parent Hamiltonian by $\bigotimes_j U_j$ preserves its gap exactly and increases interaction range by only a bounded amount.

The unknown state is promised to be

$$\rho_c(U) = U|c\rangle\langle c|U^\dagger, \qquad U = \bigotimes_j U_j, \quad c \in \{C, D\}. \tag{5}$$

The nuisance U is fixed for all copies of that state. The classifier receives measurement outcomes, not U, a wavefunction, or hidden class information. The Bayes comparison uses equal class priors and independent Haar-distributed O_j sampled *once* per state. The sector measurement will have the same law for every U, so its performance also holds uniformly over the supported nuisance.

The promise fixes block alignment and restricts nuisance gates to lie within blocks. It does not cover arbitrary symmetric circuits, unknown blocking, cross-block deformations, or noisy preparations. Those changes can preserve a thermodynamic phase while changing the likelihood used here.

3 Purities and joint block-SWAP laws

For $A \subseteq \{0, \ldots, M-1\}$, let $\rho_{c,A}$ be the state reduced to the union of blocks indexed by A. Write $b(A)$ for the number of nearest-neighbor ring edges joining a selected block to an unselected block.

Proposition 1 (Matched block purities)**.** *For both representatives, and for every supported nuisance,*

$$\operatorname{Tr} \rho_{C,A}^2 = \operatorname{Tr} \rho_{D,A}^2 = 2^{-b(A)}. \tag{6}$$

Consequently, measuring all block SWAPs on two copies gives identical joint probability laws in the two classes. Any number of independent repetitions of that block-SWAP-only experiment has equal-prior classification accuracy $1/2$*.*

Proof. For the dimer state, each cut interblock bond severs one maximally entangled pair, contributing a factor $1/2$ to purity. For a graph state, the Schmidt rank across a cut is 2^r, with a flat nonzero spectrum, where r is the rank over $\mathbb{F}_2$ of its cut adjacency matrix. Here the crossing cluster edges form a matching: each block contains two sites, so distinct boundary edges have distinct endpoints. Their cut-matrix rows and columns are independent, giving $r = b(A)$. Block-local unitaries preserve these purities.

Let F_j swap block j between copies, with outcomes $f_j \in \{-1, 1\}$. For every A, $\mathbb{E} \prod_{j \in A} f_j = \operatorname{Tr} \rho_{c,A}^2$. These 2^M moments determine the entire law through

$$\Pr_c(f) = 2^{-M} \sum_A \left(\prod_{j \in A} f_j \right) \operatorname{Tr} \rho_{c,A}^2. \tag{7}$$

The two laws therefore coincide, and their repeated-product laws coincide as well. □

This is stronger than matching a single half-chain entropy. It also shows why a successful purity classifier on product-trivial versus cluster data would be inconclusive here: the entangled trivial representative removes that simple cue. The proposition concerns unions of the declared whole blocks. It makes no assertion that all possible sub-qubit entanglement diagnostics coincide.

4 Four sectors from an ordinary Bell circuit

Work first in the four-dimensional real basis of Eq. (4). On a pair of corresponding blocks, let F swap the copies and define

$$|\Omega\rangle = \frac{1}{2} \sum_{i=0}^{3} |ii\rangle, \quad P_0 = |\Omega\rangle\langle\Omega|, \quad J_{ij,kl} = \frac{1}{2} \epsilon_{ijkl}. \tag{8}$$

Here $\epsilon_{0123} = 1$ and is completely antisymmetric. The four projectors are

$$\begin{aligned} E_0 &= P_0, & E_9 &= (\mathbb{I} + F)/2 - P_0, \\ E_+ &= (\mathbb{I} - F)/4 + J/2, & E_- &= (\mathbb{I} - F)/4 - J/2. \end{aligned} \tag{9}$$

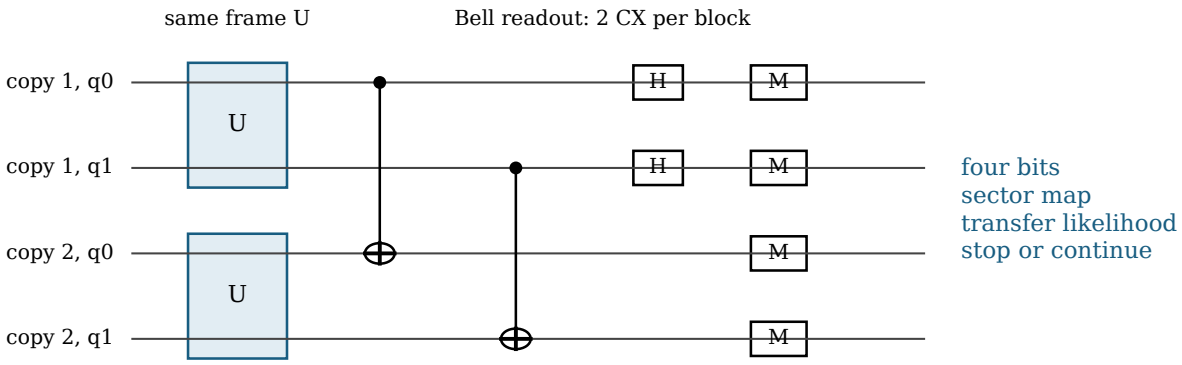


Figure 1: Native Bell readout for one two-qubit block. The repeated U denotes the same unknown preparation frame in both copies, not a gate supplied to the classifier. Each copy may be entangled with other blocks. Two CX gates and two Hadamards produce four measured bits; Eq. (11) gives the sector and Eq. (14) gives the complete-record likelihood. Optional readout twirls are omitted.

They have ranks $d = (1, 9, 3, 3)$. Direct multiplication gives $J^2 = (\mathbb{I} - F)/2$, $FJ = -J$, and orthogonality and completeness of the four effects. They resolve the trace, symmetric-traceless, self-dual antisymmetric and anti-self-dual antisymmetric components of $\mathbb{C}^4 \otimes \mathbb{C}^4$. Each component is irreducible under $O \otimes O$, $O \in \mathrm{SO}(4)$, and occurs once. One way to see the splitting is $\mathrm{Spin}(4) \simeq \mathrm{SU}(2) \times \mathrm{SU}(2)$: the vector tensor square decomposes as $(1,1) \oplus (3,3) \oplus (3,1) \oplus (1,3)$. The commutant therefore has dimension four.

The Hodge splitting is exceptional to local dimension four. An orthogonal transformation with determinant -1 exchanges E_+ and E_-. A full $O(4)$ average would erase this distinction; an $\mathrm{SO}(4)$ average retains it. The usual real-Clifford orthogonal-design result [13] does not by itself establish the orientation-preserving twirl required below.

Transform Eq. (9) to the physical basis using $V \otimes V$. The resulting projectors are diagonal in the ordinary transversal Bell basis. For each physical site apply

$$\mathrm{CX}(\text{copy } 1, \text{copy } 2), \qquad H(\text{copy } 1), \tag{10}$$

then measure both copies. Let z_i be the first-copy output and x_i the second-copy output. For block j use little-endian integers $x^{(j)} = x_{2j} + 2x_{2j+1}$ and $z^{(j)} = z_{2j} + 2z_{2j+1}$. The sector is $s_j = f(4x^{(j)} + z^{(j)})$, where, in order $0, \ldots, 15$,

$$\begin{aligned} f = [&1,1,1,1,1,3,1,3,\ 1,1,2,2,0,2,3,1], \\ &(0,1,2,3) \equiv (0,9,+,-). \end{aligned} \tag{11}$$

There are N Bell CX gates, N Hadamards and $2N$ measured bits per pair of copies. Resolving four sectors costs no additional ideal quantum gate compared with retaining only block-SWAP signs. It changes classical decoding of the same bitstrings. Since the effects commute with every $O_j \otimes O_j$, the complete joint sector law is invariant under every nuisance in Eq. (5).

5 Exact transfer likelihood

For a real stabilizer state, transversal Bell probabilities are 2^{-N} on its stabilizer Pauli support and zero elsewhere [6]. The stabilizer signs affect amplitudes but not these probabilities. In the cluster

Retained information	Two-copy success probability
Block SWAP only	1/2
Trace sector and SWAP	$41/64 = 0.640625$
Chirality and SWAP	$21/32 = 0.65625$
All four sectors	3/4

Table 1: Exact coarsenings of the same noiseless Bell measurement, with four two-qubit blocks and equal class priors. "Chirality and SWAP" retains the two antisymmetric sectors but merges the symmetric sectors. These are fixed-budget Bayes probabilities, not empirical accuracies or stopping guarantees.

case the support satisfies $z_i = x_{i-1} \oplus x_{i+1}$ with uniform x; in the dimer case x and z are independently equal along each dimer bond.

These constraints yield four nonnegative 4×4 matrices for each class. Write $p = p_0 + 2p_1$ and $q = q_0 + 2q_1$, with $p_i, q_i \in \{0, 1\}$, and define

$$\begin{aligned} x_C(p,q) &= p_1 + 2q_0, & z_C(p,q) &= (p_0 \oplus q_0) + 2(p_1 \oplus q_1), \\ x_D(p,q) &= p_0 + 2q_0, & z_D(p,q) &= p_1 + 2q_1, \end{aligned} \tag{12}$$

where $\oplus$ is addition modulo two. Put

$$T_{c,s}[p,q] = \mathbf{1}\{f(4x_c(p,q) + z_c(p,q)) = s\}. \tag{13}$$

For the complete ring record $s = (s_0, \ldots, s_{M-1})$,

$$p_c(s) = 4^{-M} \operatorname{Tr}\left(\prod_{j=0}^{M-1} T_{c,s_j} \right). \tag{14}$$

Indeed, expanding the trace sums over a periodic hidden string $p_0, \ldots, p_{M-1}$. Each hidden string has weight 4^{-M}, and its local constraints give precisely the Bell support above. Summing the sector indicators gives Eq. (14).

The matrix size is independent of M, so evaluating one likelihood costs $O(M)$ arithmetic operations. The implementation uses exact integer numerators for moderate M and rescaled nonnegative products for log likelihoods. It never enumerates a test-state wavefunction during inference. A separate analytic record sampler is a simulator of Eq. (14), not evidence of gate-level circuit execution.

For $M = 4$, finite enumeration of the 4^4 records gives the following equal-prior success probabilities: The full distributions satisfy

$$\frac{1}{2} \sum_s |p_C(s) - p_D(s)| = \frac{1}{2}. \tag{15}$$

A direct integer check multiplies Eq. (14) by 256 before summation. The dimer-only support has probability 1/2 under D and zero under C. On their common support, $p_C \geq p_D$. Thus the decision "dimer on dimer-only support; cluster otherwise" already attains 3/4.

6 An exact two-copy separation

The averaged two-copy inputs are

$$\overline{\rho}_c^{(2)} = \int \rho_c(U)^{\otimes 2} \, dU. \tag{16}$$

The same nuisance acts on the two copies inside the integral. Replacing this with independently randomized copies defines a different problem.

Theorem 2 (Two-copy sector and PPT separation)**.** *For $M = 4$ and Eq. (16), unrestricted two-copy Bayes success is exactly $3/4$ and is attained by the sector measurement. Every two-outcome POVM whose effects have positive partial transpose across the partition between the complete copies has success probability at most*

$$\frac{94162839433905}{140737488355328} < \frac{67}{100} < \frac{3}{4}. \tag{17}$$

In particular, this bound holds for adaptive measurements of two separate copies with only classical communication and no shared entangled resource between copies. The sector protocol succeeds with probability $3/4$ for every supported nuisance.

Proof. Average any effect over the product of the blockwise SO(4) groups acting identically on the two copies. This does not change its performance on Eq. (16). Multiplicity-freeness implies that the averaged effect has the form

$$E = \sum_{s\in\{0,9,+,-\}^M} u_s \bigotimes_j E_{s_j}, \qquad 0 \le u_s \le 1. \tag{18}$$

Optimizing independently over u_s is classical likelihood discrimination. Equation (15) gives unrestricted success $3/4$.

Let Γ transpose the second copy in the real basis. Local calculation gives $F^\Gamma = 4P_0$, $P_0^\Gamma = F/4$, and $J^\Gamma = -J$. In the order $(0, 9, +, -)$, coefficients transform by

$$C = \frac{1}{4}\begin{pmatrix} 1 & 9 & -3 & -3 \\ 1 & 1 & 1 & 1 \\ -1 & 3 & -1 & 3 \\ -1 & 3 & 3 & -1 \end{pmatrix}, \qquad C^2 = \mathbb{I}. \tag{19}$$

PPT is preserved by the twirl, whose rotations are local relative to the copy cut. Write $G = C^{\otimes M}$ and $\delta = p_C - p_D$. Positivity of both E^Γ and $(\mathbb{I} - E)^\Gamma$ is exactly $0 \le Gu \le 1$. PPT success is therefore

$$\frac{1}{2} + \frac{1}{2}\max\{\delta^\mathsf{T} u : 0 \le u \le 1,\ 0 \le Gu \le 1\}. \tag{20}$$

This is a 256-variable linear program for $M = 4$.

The upper bound uses exact weak duality, rather than numerical feasibility. Set $A = [G; -G]$, $b = [\mathbf{1}; \mathbf{0}]$. Any nonnegative rational λ yields

$$\delta^\mathsf{T} u \le \lambda^\mathsf{T} b + \sum_i \max\{\delta_i - (A^\mathsf{T}\lambda)_i, 0\}. \tag{21}$$

To obtain it, add $\lambda^\mathsf{T}(b - Au) \ge 0$ and maximize each residual coefficient over $0 \le u_i \le 1$. The supplied certificate has 512 nonnegative integer numerators with common denominator 2^{40}. Both G and δ have denominator 256. Substituting those numerators into Eq. (21) with integer arithmetic gives Eq. (17). The enumeration and integer verification are specified in Appendix B.

A protocol that measures the copies locally and communicates classical outcomes has separable effects across the copy cut, hence PPT effects [11]. This includes arbitrary entangling operations within either copy and classical randomization. Finally, sector invariance makes its law independent of U, proving the uniform performance statement. □

The certificate is slightly above the floating-point optimum 0.6690672153635. Its validity does not depend on a solver's tolerance or a rounded primal solution. The gap is a restriction on two-copy measurement access. It is not a lower bound for protocols with more total copies, a computational-complexity separation, or evidence that every single-copy circuit is inferior at every budget.

6.1 An orientation-robust extension

Enlarge the block nuisance from SO(4) to $O(4)$. Reflections exchange E_+ and E_-, so merge them into $E_{\mathrm{a}} = E_+ + E_-$. The three ranks are then $(1, 9, 6)$. This coarsening is invariant under reflections and has exact two-copy success 41/64. In this basis the partial-transpose coefficient matrix is

$$C_O = \frac{1}{4}\begin{pmatrix} 1 & 9 & -6 \\ 1 & 1 & 2 \\ -1 & 3 & 2 \end{pmatrix}. \tag{22}$$

Replacing C by C_O and summing the two antisymmetric probabilities reduces Eq. (20) to 81 variables. An independently reconstructed integer weak-duality certificate gives

$$P_{\mathrm{PPT}}^{O(4)} \le \frac{169324790679249}{281474976710656} \simeq 0.6015625000062 < \frac{41}{64}. \tag{23}$$

These block gates preserve the virtual time-reversal indices in Appendix A. A connected symmetric path can also be obtained after adjoining one trivial real ancilla per block: $O \otimes \mathbb{I}_2 \in \mathrm{SO}(8)$, and a real orthogonal path leaves that ancilla unchanged at its endpoint. This stable-equivalence argument does not require ancillas in the measurement circuit. Thus a smaller two-copy gap survives the enlarged nuisance family. The four-sector likelihood is not reflection-invariant and must not be applied unchanged to that family. This is an extension of the orbit-discrimination promise, not robustness to arbitrary cross-block symmetric circuits.

7 Repeated pairs and stopping

For repeated sector measurements of a fixed supported state, each pair consumes two fresh copies. Invariance gives iid records with law p_c, even though the hidden nuisance is never redrawn. The Bhattacharyya coefficient at $M = 4$ is

$$B = \sum_s \sqrt{p_C(s)p_D(s)} \simeq 0.6884105351. \tag{24}$$

For r independent pair measurements, the optimal fixed-budget Bayes error obeys

$$P_{\mathrm{err}}^{(r)} = \frac{1}{2}\sum_{s_1,\ldots,s_r} \min\left\{\prod_k p_C(s_k), \prod_k p_D(s_k)\right\} \le \frac{1}{2}B^r. \tag{25}$$

Eleven pair measurements, or 22 copies, make this bound less than 1%. This is an upper bound for one specified protocol, with no matching many-copy lower bound for separate-copy methods.

A sequential binary rule can instead accumulate

$$L_r = \sum_{k=1}^{r} \log\frac{p_C(s_k)}{p_D(s_k)}. \tag{26}$$

Accept C at $L_r \ge \log(1/\alpha)$, accept D at $L_r \le -\log(1/\alpha)$, and otherwise continue or abstain at a cap. Under D, $\exp L_r$ is a nonnegative likelihood-ratio supermartingale; under C the reverse ratio is one. Missing support gives the standard extended-real likelihood convention. The maximal inequality therefore bounds the probability of ever accepting the wrong class by α under either supported class. This is the standard likelihood-ratio mechanism, with quantum sequential universal tests, quantum confidence sequences and continuous-parameter sequential tests providing related established frameworks [14–16].

The guarantee requires the exact supported laws, stationary fresh copies and the stated measurement model. It bounds wrong acceptance over all trajectories, not error conditional on acceptance. Runtime log likelihoods use binary64 arithmetic and are independently cross-checked; the implementation is not a formal roundoff certificate. The state $\overline{\rho}_c^{(2r)}$ with persistent nuisance is generally not $(\overline{\rho}_c^{(2)})^{\otimes r}$. Consequently, the two-copy PPT bound cannot be exponentiated into a many-pair lower bound. Only the invariant sector record law factorizes as used above.

8 Readout noise and finite twirling

A classical readout channel can distinguish Bell basis vectors inside the same ideal sector. Fixed-nuisance invariance then need not survive. For known independent flips of each measured bit with probability η, let D_y be the noisy local effect for reported sector y.

In the real basis, close the group generated by orientation-preserving signed permutations and $H \otimes H$. Exact half-integer arithmetic gives a finite subgroup $\mathcal{G} \subset \mathrm{SO}(4)$ of order 576. Its tensor-square commutant dimension follows from

$$\frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} (\operatorname{Tr} g)^4 = 4. \tag{27}$$

The four-dimensional SO(4) commutant is contained in this finite-group commutant; equality of dimensions makes the two spaces equal. Their Hilbert–Schmidt orthogonal projections, and hence two-copy twirls, are equal. Signed permutations alone have commutant dimension five and fail this check. The physical gates $VgV^\dagger$ are Clifford gates. This is an explicit orientation-preserving finite twirl, not a claim to invent Clifford designs [13].

Apply a fresh uniform g identically to both copies of each block before Bell readout, and discard its classical setting. Schur averaging gives

$$\mathcal{T}(D_y) = \sum_s K_{ys} E_s, \qquad K_{ys} = \frac{\operatorname{Tr}(E_s D_y)}{d_s}. \tag{28}$$

Equivalently, average the bit-flip probability of reporting y uniformly over the d_s Bell states in sector s. The noisy likelihood is still Eq. (14), with

$$\widetilde{T}_{c,y} = \sum_s K_{ys} T_{c,s}. \tag{29}$$

The preparation nuisance remains fixed. The twirl is fresh measurement randomization.

For the Bell-diagonal readout effects, the implementation reduces the randomization to nine settings. Their induced Bell permutations form a group transitive on each sector. Uniform averaging therefore replaces every diagonal effect by its sector average, which proves Eq. (28) for this readout channel. This nine-setting statement does not assert an exact twirl of arbitrary nondiagonal effects.

The recorded logical synthesis uses all-to-all connectivity and optimization level 1, with basis gates R_z, $\sqrt{X}$, X and CX. Of the 576 elements, 64 require zero CX gates per block per copy, 256 require one and 256 require two. The nine readout settings comprise one zero-CX setting, four one-CX settings and four two-CX settings. Both constructions have mean $4/3$ and maximum two. The readout twirl therefore adds $(8/3)M$ expected CX gates per pair, in addition to the $2M$ Bell CX gates.

8.1 A specified gate-noise correction

The noisy-twirl experiment assumes that each twirl CX is followed by complete-block depolarization with probability γ, separately on each copy. All single-qubit gates, preparation and Bell CX gates

Eight-qubit condition	Copies	Coverage	Wrong/all	Bell CX	Twirl CX
Ideal	6.3485	1.00000	51/4000	25.394	0
Readout 0.03	10.7485	0.99975	126/4000	42.994	57.325
Readout 0.03, twirl $\gamma = 0.03$	15.5975	0.99750	126/4000	62.390	84.858

Table 2: Simulated sequential tests retaining every sector in each pairshot. Mean copies charge the cap to abstentions. Coverage is the fraction with an accepted label; wrong/all is unconditional wrong acceptance. CX columns exclude preparation and routing; twirl counts are expectations under the randomized setting law. The last row uses whole-block depolarization after twirl CX gates only.

are ideal in this model. For setting j with c_j twirl CX gates per copy, put $v_j = (1-\gamma)^{2c_j}$. The Bell-diagonal noisy effects have maximally mixed one-copy marginals; applying these channels therefore gives

$$D_{y,j}^{\text{noisy}} = v_j D_{y,j} + (1 - v_j) a_y \mathbb{I}, \qquad a_y = \operatorname{Tr}(D_y)/16 = d_y/16. \tag{30}$$

Choose the setting with probability $\pi_j = v_j^{-1} / \sum_k v_k^{-1}$. Since $\pi_j v_j$ is constant, averaging gives

$$D_y^{\text{eff}} = h\mathcal{T}(D_y) + (1-h) a_y \mathbb{I}, \qquad h = \frac{9}{\sum_j v_j^{-1}}. \tag{31}$$

The effective confusion matrix is $K_{ys}^{\text{eff}} = hK_{ys} + (1-h)a_y$. At $\gamma = 0.03$, $h = 0.921231823$ and the expected twirl CX count is 1.360127852 per block per copy. This inverse-visibility sampling identity is conditional on the stated depolarizing channel. Uniform setting selection need not restore invariance when the gate-dependent visibilities differ. Generic coherent, correlated, preparation and routing errors are outside this calculation. Resource-aware importance sampling and noise tailoring have substantial prior work [17, 18]; Eq. (31) specifies a positive-channel covariance identity for this finite readout library.

9 Computational evidence and resource accounting

The exact likelihood enumeration, certificate and native measurement checks are separate evidence types. Integer arithmetic establishes Eq. (17). Numerical statevector and circuit checks verify the mapping between those formulas and the implementation; they do not replace the algebraic argument.

For eight qubits, the transfer probabilities agree with independently generated native Bell probabilities to 1.2×10^{-16}. Four independent SO(4) draws per representative preserve the joint law within 5.3×10^{-15}. Explicit local projector products and partial transpose give zero arithmetic discrepancy in the saved check. The physically conjugated finite-group noisy effects agree with Eq. (28) within 2.3×10^{-15} at readout-flip rate 0.03. All 576 physical group elements pass the recorded Clifford check.

The finite-shot protocol fixes $\alpha = 0.05$, a cap of 32 pairs, copy budgets $\{2, 4, 8, 16, 32, 64\}$, $M \in \{3, 4, 6\}$, and readout-flip rates $\{0, 0.01, 0.03, 0.05\}$. Each condition has 2000 independent trajectories per class, plus a condition with Eq. (31) at $\gamma = 0.03$. All records are generated from the analytic sector law. Because this law is nuisance-invariant, these trials assess shot randomness; they do not constitute additional distinct nuisance-state validation. Descriptive Wilson intervals for trajectory errors and all fixed-budget coarsenings are retained with the raw records.

The full-sector fixed measurement is a strong control for adaptive subset acquisition. In the ideal eight-qubit condition, cost-normalized one-step Bayes-risk acquisition uses 7.3605 mean copies and 22.0815 Bell CX gates; the full measurement uses fewer copies but more Bell CX gates. Under readout 0.03, risk acquisition uses 21.33 copies at coverage 0.813, versus 10.7485 copies and coverage 0.99975

for the full measurement. Its 97 versus 126 wrong acceptances also reflect the larger abstention fraction. Adaptive selection does not establish a copy saving in these comparisons.[1]

A separate locked experiment compares inverse-visibility sampling with uniform settings followed by deliberate visibility equalization to the worst setting, $v_{\min} = 0.88529281$. Equalization uses a randomized local Pauli channel, so it adds single-qubit operations but no CX gates in the stated model. Both methods restore the specified covariance exactly. Equalization uses independent extra Pauli draws on the two copies; sharing these draws would implement a different channel. On 8000 paired analytic-law trajectories at readout and twirl-depolarization probabilities 0.03, inverse-visibility sampling uses 15.56825 mean copies versus 18.18725. The paired mean difference is -2.619 copies, with a 2000-replicate trajectory-bootstrap 95% interval $[-2.7893, -2.4330]$. Wrong-acceptance fractions are 253/8000 and 238/8000, and coverages 0.99750 and 0.992375. Expected Bell-plus-twirl CX counts are respectively 146.972 and 169.748, excluding preparation. This demonstrates a channel-specific improvement over an explicit exact-covariance control; it is not an optimization over all noisy measurements.

An exploratory covariance-constrained optimization over the verified library's 18 distinct Bell permutations gives visibility 0.921604008 and mean 1.347005344 twirl CX gates per block per copy. The corresponding physical noisy effects match the optimized confusion model within 1.8×10^{-15}. Reusing the retained paired randomness gives 15.55925 mean copies, 254/8000 wrong acceptances and coverage 0.9975. Its paired copy difference from the nine-setting construction is -0.009, with bootstrap interval $[-0.025, 0.00725]$; no copy reduction is supported. Expected total measurement CX falls to 146.071, about 0.6% below the nine-setting rule. The simple rule is therefore not claimed to minimize resource cost over the larger library.

Native validation uses 32 independently prepared eight-qubit states, with the nuisance frame held fixed for three trajectories each. The maximum probability discrepancy from the transfer law is 1.12×10^{-15}. The 96 native-record trajectories have one wrong acceptance, complete coverage and mean 6.5625 copies. Four additional local Aer jobs, two per class, each execute 8192 shots on 16 qubits, retaining counts and circuit exports. The compiled cluster circuits use 40 CX gates per pair, comprising 32 preparation and eight Bell CX gates; the dimer circuits use 32, comprising 24 preparation and eight Bell CX gates. Logical depths are respectively 48 and 20 in the recorded all-to-all basis. These four jobs contain 32,768 simulator pairshots; five additional command-line smoke-test pairshots bring the continuation's local Aer total to 32,773. None is a physical-device experiment.

9.1 Matched held-out single-copy comparisons

An independently implemented experiment uses 160 training, 80 calibration and 200 held-out physical states per class. Each state has an independent blockwise Haar-SO(4) nuisance, fixed across its copies. Physical-state seeds are disjoint across all splits. Measurement and classical-model choices are frozen on calibration before any held-out states or outcomes are generated. Ten independent finite-shot trajectories per held-out state give 4000 trajectories per measurement arm; uncertainty is bootstrapped over the 400 physical states, stratified by class, rather than treating trajectories as independent physical examples.

The four single-copy arms are repeated all-X measurement; repeated pairs of all-X, all-Y and all-Z settings; independent uniformly random Pauli axes at each site; and a trained two-layer entangling circuit with full eight-bit readout. Classical candidates retain full outcome histograms where applicable, individual Pauli means and unbiased squares, shadow window-purity estimates, joint block-subset collision moments, and full-record likelihoods integrated over the training-state ensemble. Radial-basis support-vector machines, extra-tree ensembles and likelihood temperatures are selected on calibration.

[1] The stable replay resolves risk/information-gain score ties within 10^{-12} by the first action. It reuses the locked original records and preserves every primary full-measurement field bitwise. Original floating-argmax outputs are retained.

Copies	Single-copy (%)	Sectors (%)	Difference (pp)
2	59.075	75.225	16.150
4	65.000	88.025	23.025
8	69.650	97.075	27.425
16	81.425	99.875	18.450
32	87.125	100.000	12.875
64	93.875	100.000	6.125

Table 3: Native Born-rule simulation on 400 held-out physical states, with ten trajectories each. The single-copy measurement and classical learner are selected only on calibration. "pp" means percentage points. At eight copies the paired state-bootstrap 95% interval for the difference is $[25.15, 29.775]$ pp. Zero observed sector errors at 32 and 64 copies do not establish zero population risk.

The circuit has 12 shared rotation parameters and 14 readout CX gates. Two exact-simulator optimization starts maximize an unbiased class second-moment contrast, each capped at 25 iterations; both improve their objectives but reach the iteration limit, so convergence is not claimed. This is an implemented variational reference, not a reproduction of a published QCNN.

The strongest calibration-selected arm is the trained circuit at two through 32 copies and repeated all-X at 64. The Bell arm executes the native doubled-state CX/H simulation on the same held-out wavefunctions and retains every measured bit. The classifier receives only decoded sectors; it never receives those wavefunctions or nuisance matrices. Table 3 and Fig. 2 report the frozen comparisons.

At eight copies, sector accuracy is 97.075%. The 95% interval from the physical-state bootstrap is $[96.525, 97.600]$%. The best tested single-copy accuracy at 64 copies is 93.875%, interval $[92.775, 94.875]$%.

The complete held-out experiment generates 1,280,000 source-copy draws across five arms, including unused prefixes; it does not claim that this total equals the copies requested by any one classifier. For balanced classes, eight-copy Bell inference uses mean 144 logical CX gates including preparation: 112 for source preparation and 32 for Bell readout. The selected eight-copy circuit baseline uses 224, while 64-copy all-X uses 896. These counts assume ideal gates and all-to-all logical connectivity. Bell readout requires two coherent copies, doubling simultaneous source qubits from eight to sixteen. The sector rule has the exact family likelihood, while these finite-trained competitors approximate their decision rules. The empirical comparison establishes an improvement over the frozen tested methods under the orbit promise, not a lower bound against all longer single-copy protocols. The random-Pauli arm retains all raw snapshots but its fitted features use windows of at most four sites and local basis/outcome frequencies; its learner does not exhaust arbitrary global snapshot processing.

A state pair contains $2N$ qubits before readout, or requires quantum storage of one copy while the second is prepared. Each pairshot costs two preparations, N Bell CX gates, N Hadamards, and $2N$ measurements; twirling adds the costs stated above. State preparation, routing, reset, single-qubit noise, idle error and classical inference time remain separate resources. The PPT theorem compares allowed measurement classes at two total copies, rather than equating their circuit depths or physical error rates.

The 180,000 sequential policy evaluations reuse 60,000 pregenerated histories containing 1,920,000 pair records, equivalent to 3,840,000 simulated candidate source-copy draws. Fixed budgets and policies reuse these records rather than generating independent physical experiments for every reported endpoint. The present evidence is classical simulation and exact finite computation. No physical processor is used. The analytic transfer sampler, statevector probabilities, finite-shot draws and gate-level noisy simulations are separately identified in the artifacts.

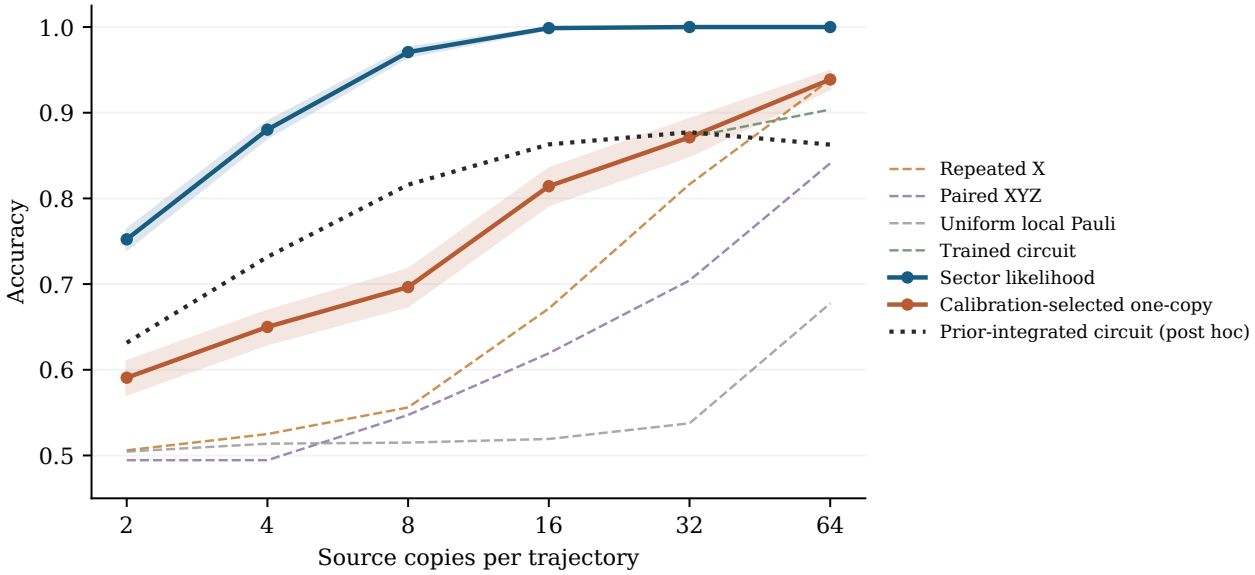


Figure 2: Matched physical-state-held-out accuracy. Solid curves with bands show sector readout and the calibration-selected single-copy competitor, with 3000-replicate stratified physical-state-bootstrap 95% intervals. Dashed curves show all four frozen single-copy arms, not test-selected replacements. The dark dotted curve is the additional prior-integrated circuit reference; its post-hoc and quadrature limitations are described below. At 32 and 64 copies all sector trajectories are correct, so the empirical bootstrap band is degenerate; it is not a population-error certificate.

9.2 A model-aware post-hoc falsifier

The exact-model versus finite-training asymmetry warrants a stronger comparison. After the locked benchmark, an additional reference is fixed without tuning on its outcomes: draw $R = 8192$ independent Haar-SO(4) nuisances per class, using fresh seeds, and approximate the known prior-integrated likelihood by

$$\widehat{\mathcal{L}}_c(y_{1:k}) = \frac{1}{R}\sum_{r=1}^{R}\prod_{t=1}^{k} p(y_t \mid c, U^{(r)}). \tag{32}$$

The nuisance is held fixed inside each product. Both the frozen entangling readout and repeated all-X are evaluated with complete existing eight-bit records, equal class priors and no likelihood-temperature tuning. This adds classical model information and computation, but no quantum measurement draws. It is a post-hoc falsification study on the already inspected cohort, not a new confirmatory endpoint.

At eight copies the model-aware circuit reference reaches 81.600%, substantially above the original selected learner's 69.650%. Sector readout remains at 97.075%: the observed difference is 15.475 percentage points, with exploratory physical-state-bootstrap 95% interval $[13.60, 17.45]$ pp. These intervals condition on the chosen quadrature and do not incorporate integration uncertainty. The effect of granting the physical model is material and narrows the empirical comparison.

The integration is not converged. Nested predeclared sizes 1024, 4096 and 8192 give circuit-reference accuracies 77.975%, 80.100% and 81.600% at eight copies, with 7% of predictions changing at the last doubling. The median importance-weight effective sample size is only 9.62 at eight copies and 1.07 at 64; a few prior draws dominate the longer-record likelihoods. At 64 copies the approximation reaches 86.275%, below the stronger original all-X learner's 93.875%. This failure cannot establish a bound on ideal model-aware inference.

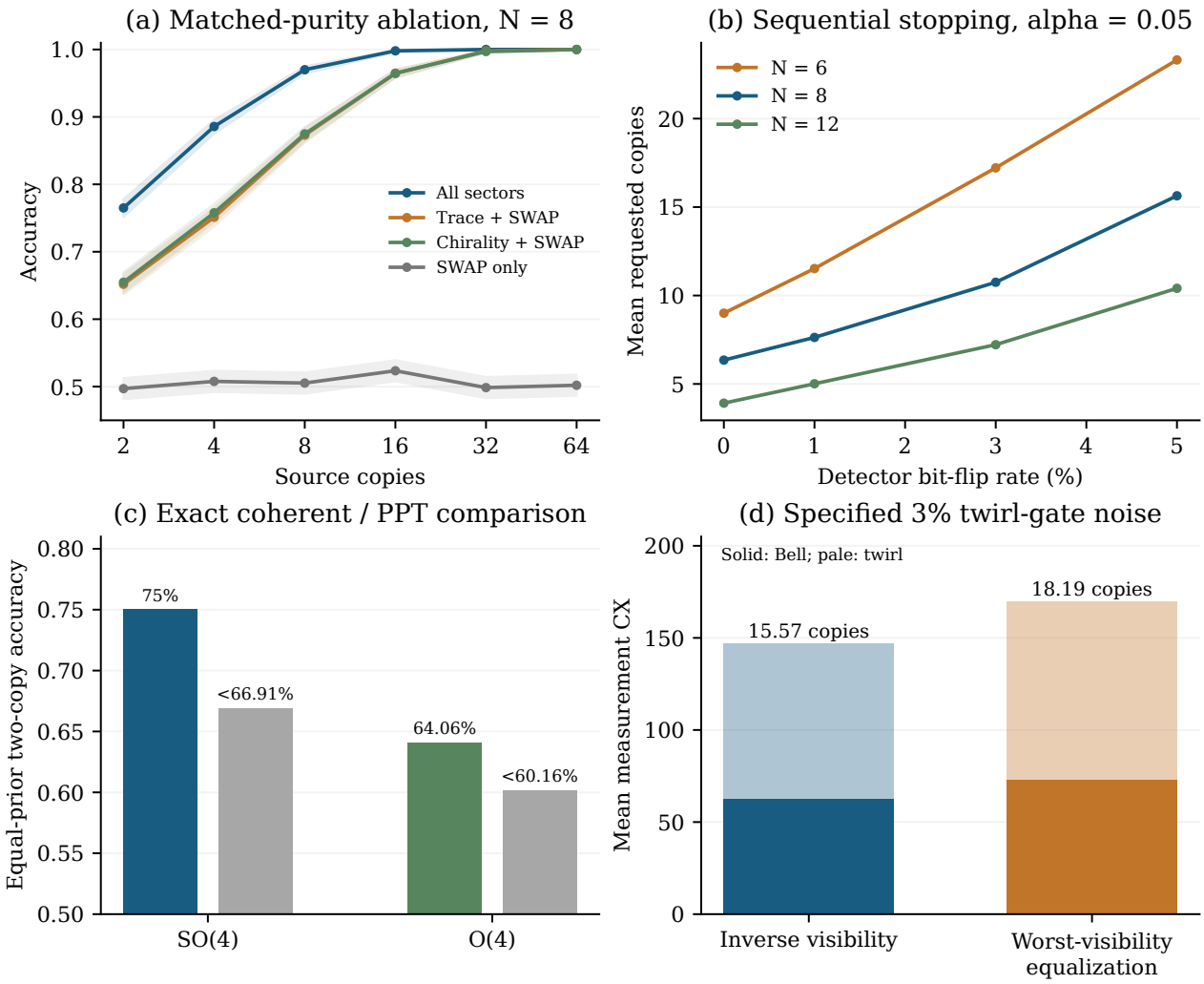


Figure 3: (a) Eight-qubit fixed-budget accuracy for four coarsenings of the same ideal Bell records, with descriptive 95% Wilson bands. (b) Mean full-sector stopping copies versus detector error with ideal-gate readout twirling, $\alpha = 0.05$ and cap 64 copies. (c) Exact two-copy coherent success and rational PPT upper bounds for the SO(4) and $O(4)$ promises. (d) Mean Bell CX (solid) and added twirl CX (pale) in the paired channel-specific noise comparison; preparation adds separate cost. Panels (a), (b) and (d) use analytic-law shot simulation; panel (c) uses exact finite computations.

For a separate exact check, the optimal two-shot repeated-all-X rule under this prior has success 113/216; the saved integration diagnostics retain this reference as well. Thus the exact two-copy PPT theorem remains the universal resource statement; optimal longer single-copy inference is unresolved.

9.3 Promise-breaking controls

Independent block reflections are outside the four-sector model but inside its $O(4)$ coarsening. On 24 reflected inputs, the unchanged SO(4) rule accepts 23 and gives nine labels inconsistent with representative ancestry. The coarsened rule accepts all 24 with one error. An added symmetric layer across the declared block cuts is unsupported by both likelihoods: the four-sector rule accepts 18 of 24 inputs with eight ancestry mismatches, and the coarse rule accepts 22 with eleven mismatches.

These counts demonstrate failure to transfer beyond the modeled orbit, not a violation of the support-conditional theorem. On maximally mixed controls, where neither clean pure-state class is the target, the rule nevertheless accepts 62 of 200 trajectories. The binary stopping test is not a general out-of-distribution detector.

10 A negative result on the original Hamiltonian benchmark

A separate retained study used three deep-phase boxes of the eight-spin model

$$H = a\sum_i Z_{i-1}X_iZ_{i+1} - b\sum_i Z_iZ_{i+1} - c\sum_i X_i, \qquad a, b, c \geq 0, \tag{33}$$

with independent thermodynamic labels and parity-sector ground states [3, 19]. It combined trained circuit readouts, first-order continuous response enclosures and support-conditional sequential testing. Perturbation expansions and quantum sequential likelihood tests are established mechanisms [14, 20].

On 120 held-out physical states and three shot trajectories per state, response intervals used 17.71 mean copies versus 23.73 for three coarse trace-distance balls, a 25.37% saving with state-bootstrap 95% interval $[23.10, 27.79]\%$. Stronger controls erased the broader efficiency claim: class-information gain used 16.36 copies, local hypothesis readouts used 16.31, and full-bitstring direct measurements correctly classified all 360 trajectories at eight fixed copies. Denser continuous trace covers also outperformed one response jet per class. The trained circuit bank did not improve the strongest acquisition bank. These are simulator observations on a narrow family, with empirical calibration and conditional validity, not a universal phase-error guarantee.

Those findings motivated the matched-purity construction; they do not become positive results under the changed promise. The original source, figures, protocols, failed superiority criterion and evidence remain in the version-1 artifact. Its state families and access model differ from Sections 2–6 and are not pooled with them.

11 Discussion

Two points survive the strongest restrictions established here. First, equal subsystem purities can coexist with distinct two-copy symmetry-sector laws for phase representatives. Second, in the specified eight-qubit nuisance problem, transversal Bell readout attains an exact success probability beyond every PPT measurement across the copy cut. The mechanism is access to invariant representation sectors that SWAP-only decoding discards.

The physical promise is essential. The detector distinguishes two blockwise symmetry orbits whose phase ancestry is known. It is not a complete invariant of arbitrary time-reversal SPT and trivial states. Cross-block symmetric deformations, unknown alignment, unmodeled preparation errors and measurement drift can invalidate its likelihood while preserving phase ancestry. Unsupported-state rejection is not supplied by the binary likelihood ratio.

The held-out experiment supports an ideal-gate improvement over the frozen tested single-copy learners within the declared orbit family. This observation and the channel-specific noise comparison do not establish an asymptotic copy advantage or optimality against arbitrary longer adaptive single-copy protocols. The finite twirl restores a specified readout symmetry at an added gate cost; its proof does not extend to generic coherent or correlated errors. Physical deployment requires independently calibrated noise, hardware mapping, coherent-copy storage and matched preparation resources.

A Independent time-reversal phase indices

The physical distinction can be checked without the proposed measurement. A bond-dimension-two matrix-product representation of the periodic cluster state is

$$A^0 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \qquad A^1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}. \tag{34}$$

Indeed $\operatorname{Tr}\prod_i A^{s_i} = 2^{-N/2}(-1)^{\sum_i s_i s_{i+1}}$, the cluster-state amplitude. With $W_C = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, these real tensors satisfy $A^{1-s} = -W_C^{-1}(A^s)^* W_C$. Blocking two sites, $B_C^{ab} = A^a A^b$, removes the scalar minus sign:

$$B_C^{1-a,1-b} = W_C^{-1}(B_C^{ab})^* W_C, \qquad W_C W_C^* = -\mathbb{I}. \tag{35}$$

For dimers, take $B_D^{ab} = |a\rangle\langle b|/\sqrt{2}$. Its ring amplitude is $\operatorname{Tr}\prod_j B_D^{a_j b_j} = 2^{-M/2}\prod_j \delta_{b_j,a_{j+1}}$, which is precisely Eq. (2). With $W_D = X$,

$$B_D^{1-a,1-b} = W_D^{-1}(B_D^{ab})^* W_D, \qquad W_D W_D^* = +\mathbb{I}. \tag{36}$$

Both blocked tensor sets span all two-by-two matrices and are injective. Their opposite virtual antiunitary-square signs give the distinct one-dimensional time-reversal indices [9]. Thus the phase labels have an independent tensor criterion, rather than being defined by the sector classifier. The retained review reconstructs both statevectors and checks these intertwining identities independently. Block-local time-reversal-preserving conjugations retain the indices along their gapped paths.

B Exact certificate and reproduction

The transfer definition in Section 5 specifies all eight integer matrices without fitting. Enumerate all 256 sector tuples with block j in base-four digit j. For each class set $n_c(s) = \operatorname{Tr}\prod_j T_{c,s_j}$, so $p_c(s) = n_c(s)/256$. Their sums are 256. The sum of absolute count differences is 256, giving Eq. (15).

For the PPT certificate define the integer matrix $A_I = 256[G; -G]$, integer vector $d_I = 256\delta$, and $L = 2^{40}$. Read the nonnegative integer vector y of length 512 from the certificate. Compute, with unbounded integers,

$$\begin{aligned} r_I &= L d_I - A_I^{\mathsf{T}} y, \\ W &= 256\sum_{j=1}^{256} y_j + \sum_i \max\{(r_I)_i, 0\}, \\ P_{\text{upper}} &= \frac{256L + W}{512L} = \frac{94162839433905}{140737488355328}. \end{aligned} \tag{37}$$

Checking $y \geq 0$ and this identity is sufficient for weak duality; no optimizer is needed to verify the certificate. The numerical linear-program optimum is a diagnostic, not the proof.

The following retained files give the definitions and evidence:

File in `research/astra-probes/`	Content
`so4_sectors.py`	Sector decoder, transfer matrices, exact and rescaled likelihoods
`isopurity_control.py`, `isopurity_control.json`	Purity masks and symmetric trivial preparation checks
`sector_likelihood.json`	Native distributions, nuisance checks and coarsenings
`transfer_and_ppt.json`	Integer certificate and local transpose checks
`verify_transfer_and_ppt.py`	Reconstructs transfer laws, numerical optimum and rational certificate
`finite_twirl.py`, `finite_twirl.json`	Exact group matrices and commutant count
`twirl_native_check.py`, `twirl_native_check.json`	Physical Clifford and noisy-effect checks
`o4_scope.py`, `o4_scope.json`	Reflection-coarsened law and rational PPT certificate
`minimal_twirl.json`, `visibility_weighting.json`	Nine settings and channel-specific weighted twirl

Recomputing a solver-derived certificate may choose different multipliers; verifying the saved integer vector preserves the stated bound. Simulator amplitudes and hidden nuisances are used only by preparation and validation code, not by the inference likelihood. The original Hamiltonian benchmark is retained separately under `paper/versions/v1/` and its corresponding evidence directories.

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